

KIRKUK, Iraq

WMO: 406210

Lat: 35.469N

Lon: 44.349E

Elev: 323

StdP: 97.50

Time Zone: 3.00 (E03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.6	(c) 3.0	(d) -10.9	(e) 1.5	(f) 11.1	(g) -8.8	(h) 1.9	(i) 13.7	(j) 10.1	(k) 12.6	(l) 8.3	(m) 12.9	(n) 1.6	(o) 60	(p) 0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.1	(c) 45.8	(d) 21.1	(e) 44.4	(f) 20.8	(g) 43.2	(h) 20.6	(i) 24.0	(j) 41.7	(k) 23.0	(l) 40.7	(m) 22.2	(n) 39.7	(o) 3.7	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.3	33.3	16.6	12.3	30.6	15.6	11.5	30.1	73.8	41.2	70.0	40.7	66.7	39.8	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.0	6.6	5.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-1.2	47.6	1.3	1.2	-2.1	48.5	-2.9	49.2	-3.6	49.9	-4.6	50.8		
			-2.1	24.5	2.1	2.8	-3.6	26.6	-4.8	28.2	-6.0	29.8	-7.5	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	9.6	11.7	15.7	21.2	28.2	34.1	36.8	32.0	25.4	16.9	12.5	(6)	
(7)		DBStd	10.16	3.06	3.30	3.49	3.85	3.63	2.55	2.13	1.96	2.98	3.78	3.23	3.17	(7)
(8)		HDD10.0	77	44	18	2	0	0	0	0	0	0	0	1	12	(8)
(9)		HDD18.3	814	270	185	97	13	0	0	0	0	1	64	183	(9)	
(10)		CDD10.0	4988	32	67	179	336	566	722	830	824	659	477	207	89	(10)
(11)		CDD18.3	2687	0	1	16	100	307	472	572	566	409	221	21	2	(11)
(12)		CDH23.3	42551	0	2	128	993	4256	8007	10181	10046	6324	2453	145	16	(12)
(13)	CDH26.7	29162	0	0	24	380	2450	5671	7704	7567	4147	1193	24	2	(13)	
(14)	Wind	WSAvg	2.3	2.1	2.2	2.4	2.5	2.7	2.7	2.5	2.2	2.1	1.9	1.9	(14)	
(15)	Precipitation	PrecAvg	476	59	60	65	69	40	12	12	8	11	32	50	59	(15)
(16)		PrecMax	634	94	114	133	122	120	37	25	18	33	82	120	119	(16)
(17)		PrecMin	285	24	14	27	23	9	3	2	2	1	6	5	13	(17)
(18)		PrecStd	84	21	23	27	29	23	9	6	4	7	20	30	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	23.2	29.2	36.9	41.1	45.1	47.7	47.2	44.1	38.0	29.1	25.2	(19)
MCWB			11.6	13.4	13.8	17.0	18.8	21.8	21.3	21.1	20.4	19.7	16.4	11.8	(20)	
2%		DB	17.9	21.0	26.2	33.2	39.1	43.8	46.0	45.8	42.3	36.1	26.9	21.8	(21)	
		MCWB	10.7	12.4	14.0	17.2	18.8	20.5	21.2	20.9	20.1	18.1	14.4	11.8	(22)	
5%		DB	16.2	19.1	24.2	30.8	37.2	42.2	44.8	44.2	40.7	34.3	24.9	19.1	(23)	
		MCWB	10.1	11.4	12.9	16.2	18.3	19.6	20.9	20.8	19.8	17.5	13.9	11.1	(24)	
10%		DB	15.0	17.6	22.2	28.8	35.8	41.1	43.6	43.1	38.9	32.8	23.0	17.5	(25)	
		MCWB	9.9	10.7	12.6	15.8	18.0	19.3	20.6	20.8	19.5	17.0	13.2	10.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	15.0	17.0	20.2	22.3	25.5	25.9	23.9	21.9	19.5	14.7	(27)	
MCD B			16.5	19.7	23.8	29.9	35.6	41.7	42.5	43.0	40.3	34.0	27.3	18.7	(28)	
2%		WB	12.5	13.9	15.7	18.9	20.9	23.1	23.7	23.8	22.1	20.4	17.5	13.8	(29)	
		MCD B	15.9	18.8	22.4	28.2	35.7	39.7	41.8	41.9	38.7	33.1	23.3	17.4	(30)	
5%		WB	11.5	12.6	14.6	17.9	19.8	21.7	22.7	23.0	21.2	19.3	16.2	13.0	(31)	
		MCD B	15.0	17.5	21.5	26.9	33.0	38.4	41.2	41.2	37.7	30.1	20.7	16.8	(32)	
10%		WB	10.5	11.5	13.7	17.0	18.9	20.5	21.8	22.2	20.5	18.3	14.7	12.1	(33)	
		MCD B	13.8	16.2	19.8	25.6	32.1	37.9	40.6	39.9	36.8	28.4	19.7	16.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	8.9	9.9	11.0	11.8	12.7	13.1	12.9	11.4	10.3	8.8	(35)	
MCDBR			9.7	10.5	12.1	13.3	13.2	13.7	13.9	13.9	14.1	12.8	12.4	11.3	(36)	
MCWBR			5.4	5.3	5.4	5.2	4.6	4.8	4.7	4.8	5.0	5.0	5.9	5.9	(37)	
5% WB		MCD BR	7.8	8.8	9.8	11.3	11.9	12.3	13.0	13.1	13.0	10.7	9.0	8.1	(38)	
		MCWBR	5.1	5.4	5.5	5.6	5.2	4.9	5.1	4.9	5.2	4.7	5.1	5.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.405	0.436	0.478	0.563	0.562	0.501	0.528	0.489	0.452	0.473	0.409	0.400	(40)	
taud		2.048	1.973	1.888	1.725	1.742	1.869	1.815	1.904	1.981	1.935	2.079	2.072	(41)		
Ebn at Noon		756	781	786	741	749	795	770	792	799	732	746	731	(42)		
Edh at Noon		130	157	186	230	229	202	212	190	167	160	125	120	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.62	3.47	4.74	5.70	6.89	8.00	7.65	6.99	5.87	4.21	3.12	2.40	(44)	
(45)		RadStd	0.17	0.31	0.38	0.32	0.36	0.32	0.20	0.19	0.25	0.31	0.31	0.27	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page